

END-OF-COURSE PROJECT · MILESTONE 1

Project Problem Statement

Submit by the end of Week 3 [23 August 23:59 hrs] · ~1 page · individual · formative (not graded) — you'll get written feedback. This seeds your End-of-Course Project (it maps to Project Rubric criteria 1 & 2).

This submission is NOT graded. However, on time non-submission counts towards incomplete End of course project.

Name: _____ Section: A Date: 11 August 2026

1 · Working title

A short, specific name for your project.

AI Scroll: A Proactive Academic Email-to-Timeline and Calendar Agent for University Students

2 · The problem, and why it matters

Who has this problem, and why is it worth solving? 2–3 sentences. (Rubric 1)

University students receive many academic emails from courses, instructors, teaching assistants, and university offices. The problem is not only that these emails are unstructured, but also that students often do not have enough time to continuously check, understand, and update information from their inbox. As a result, important deadlines, course requirements, class changes, and submission instructions may be missed, especially when related information is spread across several emails.

3 · Who it's for, and the domain

Your user / audience and the industry (e.g. banking, healthcare, retail, your employer).

The primary users are university students who manage multiple courses, assignments, projects, and administrative requirements through email. The domain is education technology, academic productivity, and AI-assisted personal task management.

4 · Why AI — and which kind? Remember, from class 1 workshops, no one size fits all

Is AI the right tool here (vs a rule or lookup)? If yes, which type — narrow ML, a foundation model, RAG, an agent? (ILO 1)

AI is appropriate because academic emails require language understanding, context tracking, and cross-email reasoning. A rule-based system may find simple keywords such as "deadline" or "assignment," but it may fail to understand implicit tasks, updated deadlines, course-specific instructions, or relationships between several emails about the same task. The project will use a foundation model for email understanding and information extraction, RAG to retrieve related previous emails and course context, and an agent workflow to classify urgency, build timelines, answer student questions, and prepare Google Calendar updates after user confirmation.

5 · Proposed approach — build vs buy or both!

What will you build (low-code and/or code)? Which layers will you own vs rent, and why? (Rubric 2, ILO 3)

I will build the academic workflow, including email classification, urgency ranking, course grouping, task extraction, cross-email merging, timeline generation, question answering, and calendar confirmation. Existing services will be used for the infrastructure layer: Gmail API or manual email import for email access, a language-model API for reasoning and extraction, and Google Calendar API for calendar synchronization. Codex may be used as a coding and development environment to help build, test, and iterate the system, while the deployed workflow will rely on scheduled automation jobs and approved API access.

6 · Data – this packs the intelligence !

What data do you need, where from, and any access / privacy / residency constraints?

The system needs selected academic emails, email metadata, sender information, course names, task descriptions, deadlines, submission instructions, and user-confirmed calendar events. The data may come from a student's Gmail inbox, a selected academic email label, or manually uploaded email text in the first version. Since emails may contain private academic and personal information, the system should request explicit user permission, process only selected academic emails, store minimal structured task data, and avoid unnecessary storage of raw email content. Each extracted task should also show the source email or evidence used by the AI.

7 • Success metric & how you'll evaluate it

One metric that ties to real value, and one check (an eval) you could actually run. (evals thread)

A key success metric is the accuracy of converting academic emails into correct and useful action items. I will evaluate the system using a small manually labelled test set, for example 50 academic emails from course-related scenarios. The evaluation will compare the system output with human-labelled answers for course identification, urgency classification, task extraction, deadline extraction, cross-email merging, and calendar-event creation. A reasonable target for the first version is 85-90% accuracy for critical task and deadline extraction, with no missed high-priority deadline in the test set.

8 • Risks, limitations & responsible use

What could go wrong (incl. silent failure), and who might it affect?

The main risk is silent failure, where the agent misses, misclassifies, or incorrectly updates an important academic task without the student noticing. This could cause missed deadlines, wrong preparation, duplicated calendar events, or confusion about course requirements. Other risks include privacy concerns, incorrect time-zone handling, over-reliance on AI, and misunderstanding updated instructions across multiple emails. To reduce these risks, the system will not directly change the student's calendar without confirmation. It will show the source email, extracted evidence, confidence level, and proposed calendar event before the user approves any update.

9 • Smallest first version (optional)

The tiniest end-to-end slice you could build first.

The smallest first version will focus on one course and a small set of academic emails. The user can upload or select several emails from the same course. The system will classify their urgency, extract tasks and deadlines, merge related updates, display a simple course timeline, answer basic questions about the emails, and create one Google Calendar event after user confirmation.

How this is used

This milestone is not graded — it's to give you fast, written feedback while there's still time to change direction. A strong problem statement makes criteria 1 (problem & significance) and 2 (business & technical trade-offs) and helps you start on a strong footing. Bring questions during class breakouts or email the course account or WA text me

INSTRUCTOR FEEDBACK

For marker use — formative comments on the problem & significance (criterion 1) and the business/technical trade-offs (criterion 2), plus one thing to strengthen before the final project.